

CLOUD SERVICE AND DEPLOYMENT MODELS

COVERAGE

- **Presenting an Analogy**
- **Comparison with Cloud**
- **Cloud Service Models**
- **Cloud Deployment Models**



IT'S ALL ABOUT THE SERVICE THAT YOU WANT

**Say, you just moved to a city
and you are looking for a
place to live.**



What could be your choices ?



Built a new house ?

Buy an empty house ?

Live in a hotel ?

Build a new house...

**You can fully
control
everything your
like your new
house to have.
But that is a
hard work ...**



If you buy an empty house ?



You can customize some part of your house. But never change the original architecture.

How about live in a hotel ?



Live in a hotel will be a good idea if the only thing you care is enjoy your life!!

There is nothing you can do with the house except living in it.

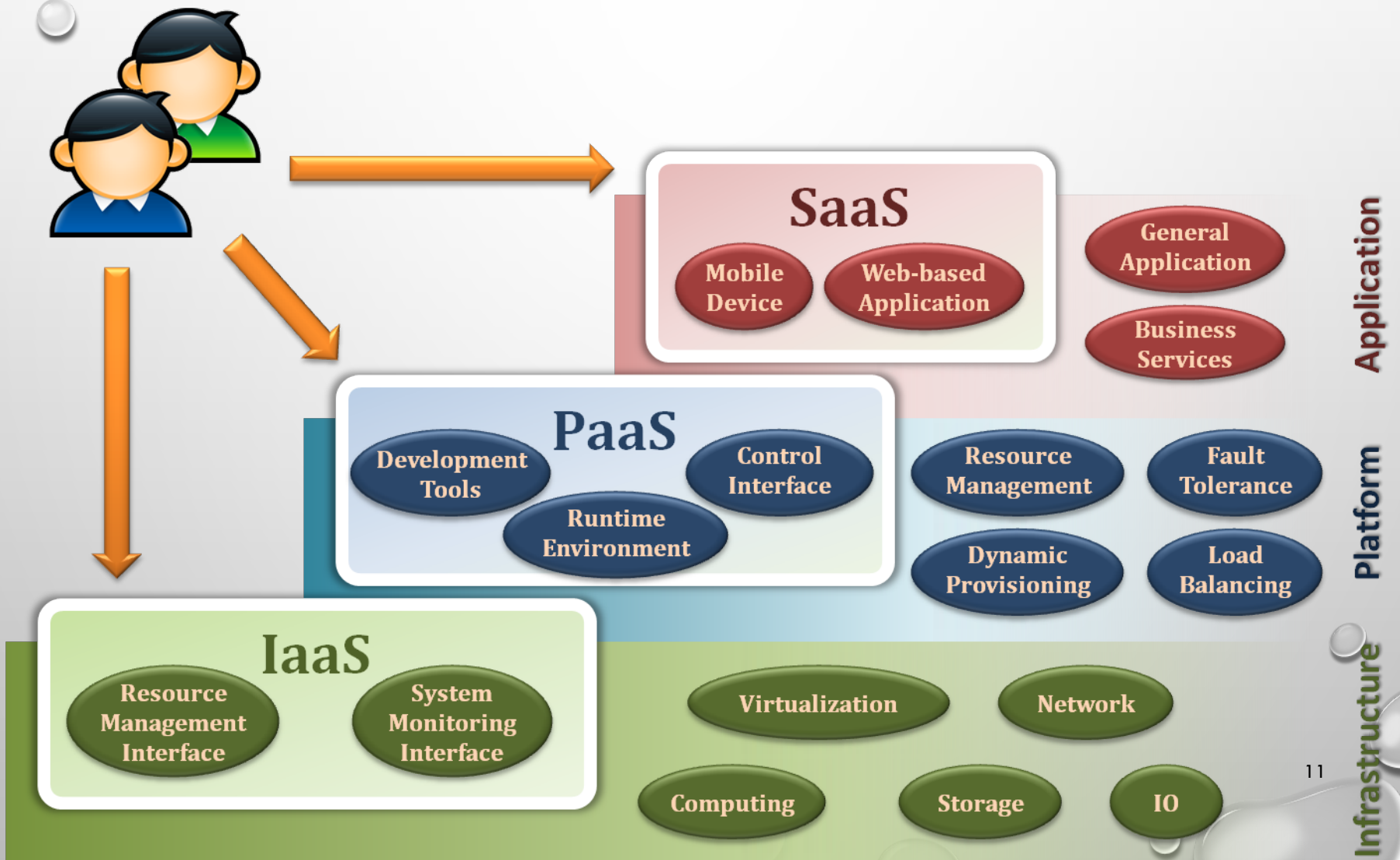
The slide features a light gray background with a subtle gradient. In the top-left and bottom-right corners, there are several realistic water droplets of varying sizes, rendered with soft shadows and highlights to give them a three-dimensional appearance. The text is centered and reads:

**Let's translate
to
Cloud Computing !!**

SERVICE MODELS OVERVIEW

- WHAT IF YOU WANT TO HAVE AN IT DEPARTMENT ?
 - SIMILAR TO **BUILD A NEW HOUSE** IN PREVIOUS ANALOGY
 - YOU CAN RENT SOME VIRTUALIZED INFRASTRUCTURE AND BUILD UP YOUR OWN IT SYSTEM AMONG THOSE RESOURCES, WHICH MAY BE FULLY CONTROLLED.
 - TECHNICAL SPEAKING, USE THE **INFRASTRUCTURE AS A SERVICE (IAAS)** SOLUTION.
 - SIMILAR TO **BUY AN EMPTY HOUSE** IN PREVIOUS ANALOGY
 - YOU CAN DIRECTLY DEVELOP YOUR IT SYSTEM THROUGH ONE CLOUD PLATFORM, AND DO NOT CARE ABOUT ANY LOWER LEVEL RESOURCE MANAGEMENT.
 - TECHNICAL SPEAKING, USE THE **PLATFORM AS A SERVICE (PAAS)** SOLUTION.
 - SIMILAR TO **LIVE IN A HOTEL** IN PREVIOUS ANALOGY
 - YOU CAN DIRECTLY USE SOME EXISTED IT SYSTEM SOLUTIONS, WHICH WERE PROVIDED BY SOME CLOUD APPLICATION SERVICE PROVIDER, WITHOUT KNOWING ANY DETAIL TECHNIQUE ABOUT HOW THESE SERVICE WAS ACHIEVED.
 - TECHNICAL SPEAKING, USE THE **SOFTWARE AS A SERVICE (SAAS)** SOLUTION.

SERVICE MODEL OVERVIEW



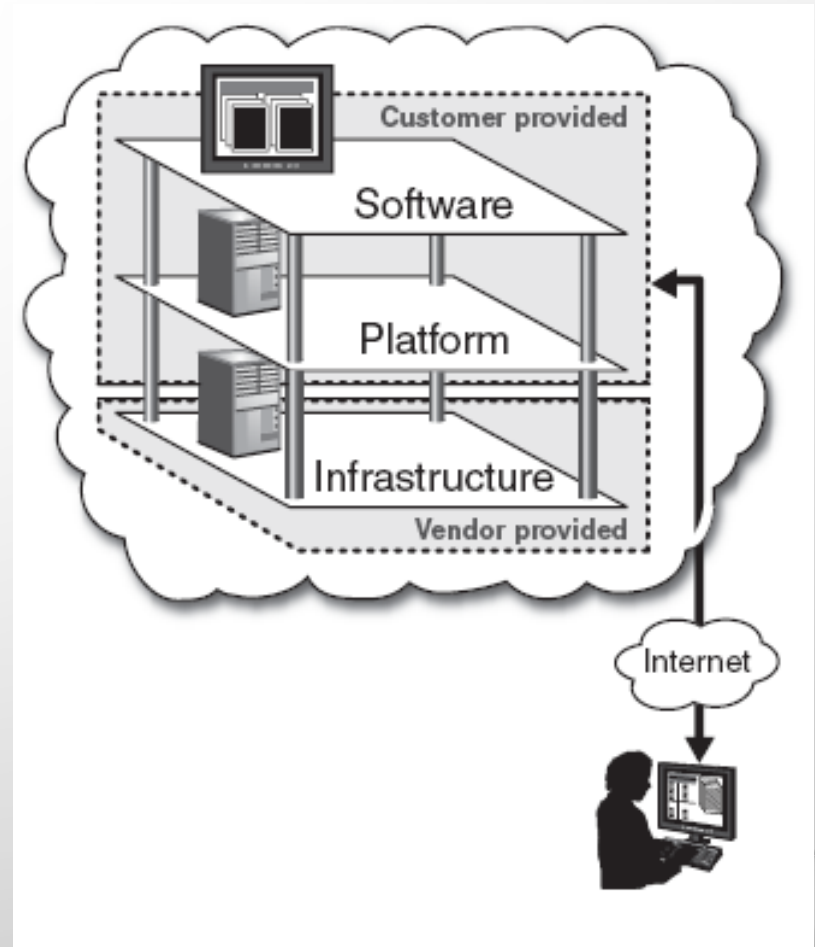
SERVICE MODELS

- INFRASTRUCTURE AS A SERVICE
- PLATFORM AS A SERVICE
- SOFTWARE AS A SERVICE

SERVICE MODELS

INFRASTRUCTURE-AS-A-SERVICE (IAAS)

- A SERVICE MODEL THAT INVOLVES OUTSOURCING THE BASIC INFRASTRUCTURE USED TO SUPPORT OPERATIONS--INCLUDING STORAGE, HARDWARE, SERVERS, AND NETWORKING COMPONENTS.
- THE SERVICE PROVIDER OWNS THE INFRASTRUCTURE EQUIPMENT AND IS RESPONSIBLE FOR HOUSING, RUNNING, AND MAINTAINING IT. THE CUSTOMER TYPICALLY PAYS ON A PER-USE BASIS.
- **THE CUSTOMER USES THEIR OWN PLATFORM (WINDOWS, UNIX), AND APPLICATIONS**

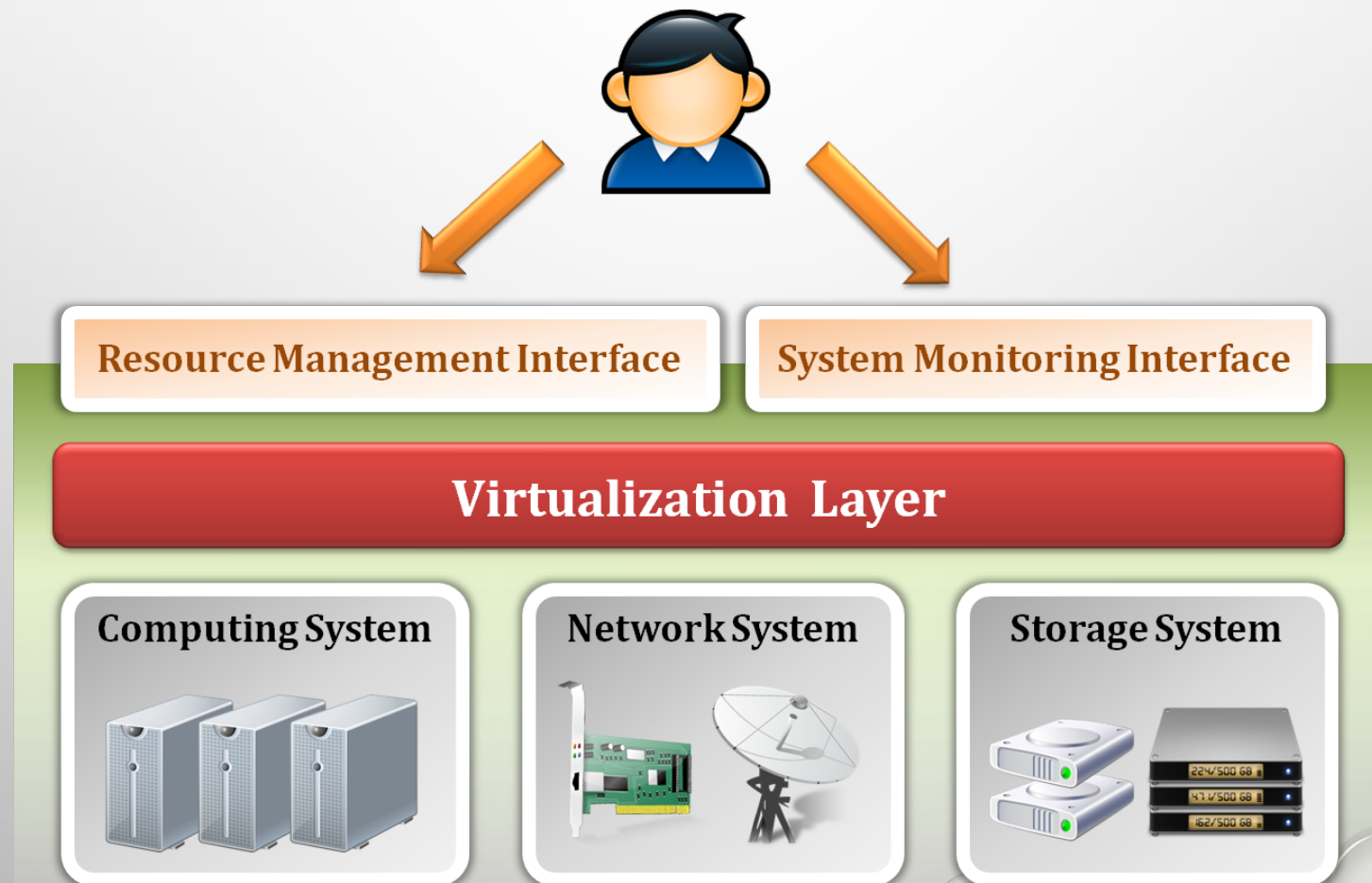


INFRASTRUCTURE AS A SERVICE

- INFRASTRUCTURE AS A SERVICE - IAAS
 - THE USER IS ABLE TO DEPLOY AND RUN ARBITRARY SOFTWARE, WHICH CAN INCLUDE OPERATING SYSTEMS AND APPLICATIONS.
 - IAAS GIVES THE CAPABILITY TO PROVISION PROCESSING, STORAGE, NETWORKS AND OTHER COMPUTE RESOURCES
 - THE CONSUMER DOES NOT MANAGE OR CONTROL THE UNDERLYING CLOUD INFRASTRUCTURE BUT HAS CONTROL OVER OPERATING SYSTEMS, STORAGE, DEPLOYED APPLICATIONS, AND POSSIBLY LIMITED CONTROL OF SELECT NETWORKING COMPONENTS .
 - SERVICES OFFERED BY THIS DELIVERY MODEL INCLUDE: SERVER HOSTING, WEB SERVERS, STORAGE, COMPUTING HARDWARE, OPERATING SYSTEMS, VIRTUAL INSTANCES, LOAD BALANCING, INTERNET ACCESS, AND BANDWIDTH PROVISIONING.
- EXAMPLES :
 - AMAZON EC2
 - EUCALYPUTS
 - OPENNEBULA..... ETC

INFRASTRUCTURE AS A SERVICE

- SYSTEM ARCHITECTURE :



INFRASTRUCTURE AS A SERVICE

- ENABLING TECHNIQUE - ***VIRTUALIZATION***
 - VIRTUALIZATION IS AN ABSTRACTION OF LOGICAL RESOURCES AWAY FROM UNDERLYING PHYSICAL RESOURCES.
 - VIRTUALIZATION TECHNIQUE SHIFT OS ONTO HYPERVISOR.
 - MULTIPLE OS SHARE THE PHYSICAL HARDWARE AND PROVIDE DIFFERENT SERVICES.
 - IMPROVE UTILIZATION, AVAILABILITY, SECURITY AND CONVENIENCE.

INFRASTRUCTURE AS A SERVICE

- PROPERTIES SUPPORTED BY VIRTUALIZATION TECHNIQUE :
 - MANAGEABILITY AND INTEROPERABILITY
 - AVAILABILITY AND RELIABILITY
 - SCALABILITY AND ELASTICITY



INFRASTRUCTURE AS A SERVICE



INFRASTRUCTURE AS A SERVICE

- PROVIDE SERVICE –**RESOURCE MANAGEMENT INTERFACE**
 - SEVERAL TYPES OF VIRTUALIZED RESOURCE :
 - **VIRTUAL MACHINE** – AS AN IAAS PROVIDER, WE SHOULD BE ABLE TO PROVIDE THE BASIC VIRTUAL MACHINE OPERATIONS, SUCH AS **CREATION**, **SUSPENSION**, **RESUMPTION** AND **TERMINATION**, ...ETC.
 - **VIRTUAL STORAGE** – AS AN IAAS PROVIDER, WE SHOULD BE ABLE TO PROVIDE THE BASIC VIRTUAL STORAGE OPERATIONS, SUCH AS **SPACE ALLOCATION**, **SPACE RELEASE**, **DATA WRITING** AND **DATA READING**, ...ETC.
 - **VIRTUAL NETWORK** – AS AN IAAS PROVIDER, WE SHOULD BE ABLE TO PROVIDE THE BASIC VIRTUAL NETWORK OPERATIONS, SUCH AS **IP ADDRESS ALLOCATION**, **DOMAIN NAME REGISTER**, **CONNECTION ESTABLISHMENT** AND **BANDWIDTH PROVISION**, ...ETC.

INFRASTRUCTURE AS A SERVICE

- PROVIDE SERVICE – **SYSTEM MONITORING INTERFACE**
 - SEVERAL TYPES OF MONITORING METRICS :
 - **VIRTUAL MACHINE** – AS AN IAAS PROVIDER, WE SHOULD BE ABLE TO MONITOR SOME SYSTEM STATES OF EACH VIRTUAL MACHINE, SUCH AS *CPU LOADING*, *MEMORY UTILIZATION*, *IO LOADING* AND *INTERNAL NETWORK LOADING*, ...ETC.
 - **VIRTUAL STORAGE** – AS AN IAAS PROVIDER, WE SHOULD BE ABLE TO MONITOR SOME STORAGE STATES OF EACH VIRTUAL STORAGE, SUCH AS *VIRTUAL SPACE UTILIZATION*, *DATA DUPLICATION* AND *STORAGE DEVICE ACCESS BANDWIDTH*, ... ETC.
 - **VIRTUAL NETWORK** – AS AN IAAS PROVIDER, WE SHOULD BE ABLE TO MONITOR SOME NETWORK STATES OF EACH VIRTUAL NETWORK, SUCH AS *VIRTUAL NETWORK BANDWIDTH*, *NETWORK CONNECTIVITY* AND *NETWORK LOAD BALANCING*, ...ETC.

IAAS - SUMMARY

- **IAAS IS THE DEPLOYMENT PLATFORM THAT ABSTRACT THE INFRASTRUCTURE.**
- IAAS ENABLING TECHNIQUE
 - VIRTUALIZATION
 - SERVER VIRTUALIZATION
 - STORAGE VIRTUALIZATION
 - NETWORK VIRTUALIZATION
- IAAS PROVIDED SERVICES
 - RESOURCE MANAGEMENT INTERFACE
 - SYSTEM MONITORING INTERFACE

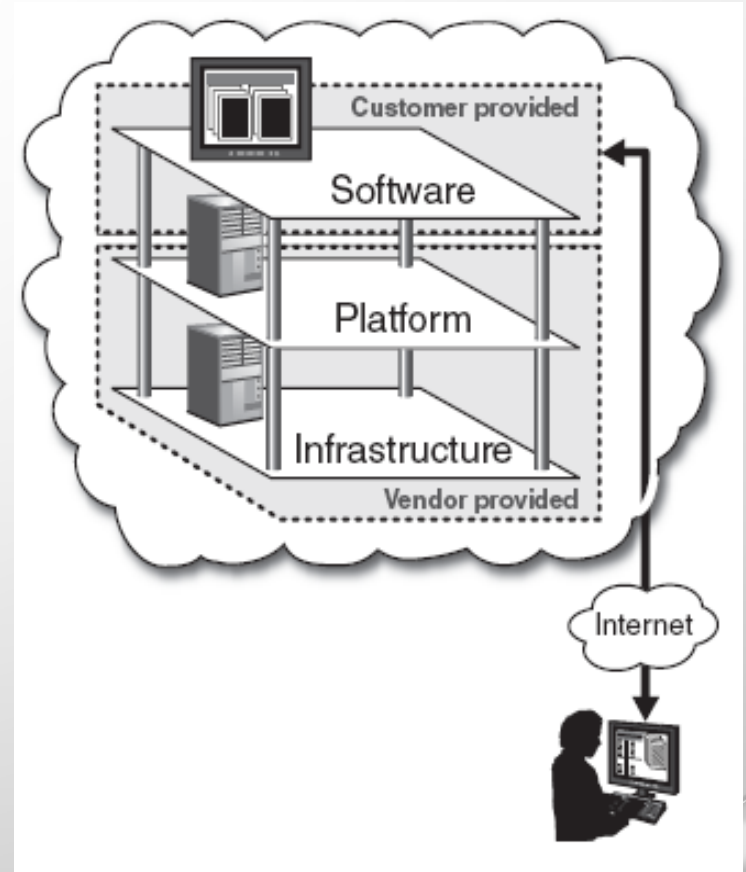
SERVICE MODELS

- INFRASTRUCTURE AS A SERVICE
- **PLATFORM AS A SERVICE**
- SOFTWARE AS A SERVICE

SERVICE MODELS

PLATFORM-AS-A-SERVICE (PAAS)

- A SERVICE MODEL THAT INVOLVES OUTSOURCING THE BASIC INFRASTRUCTURE AND PLATFORM (WINDOWS, UNIX)
- PAAS FACILITATES DEPLOYING APPLICATIONS WITHOUT THE COST AND COMPLEXITY OF BUYING AND MANAGING THE UNDERLYING HARDWARE AND SOFTWARE WHERE THE APPLICATIONS ARE HOSTED.
- **THE CUSTOMER USES THEIR OWN APPLICATIONS**

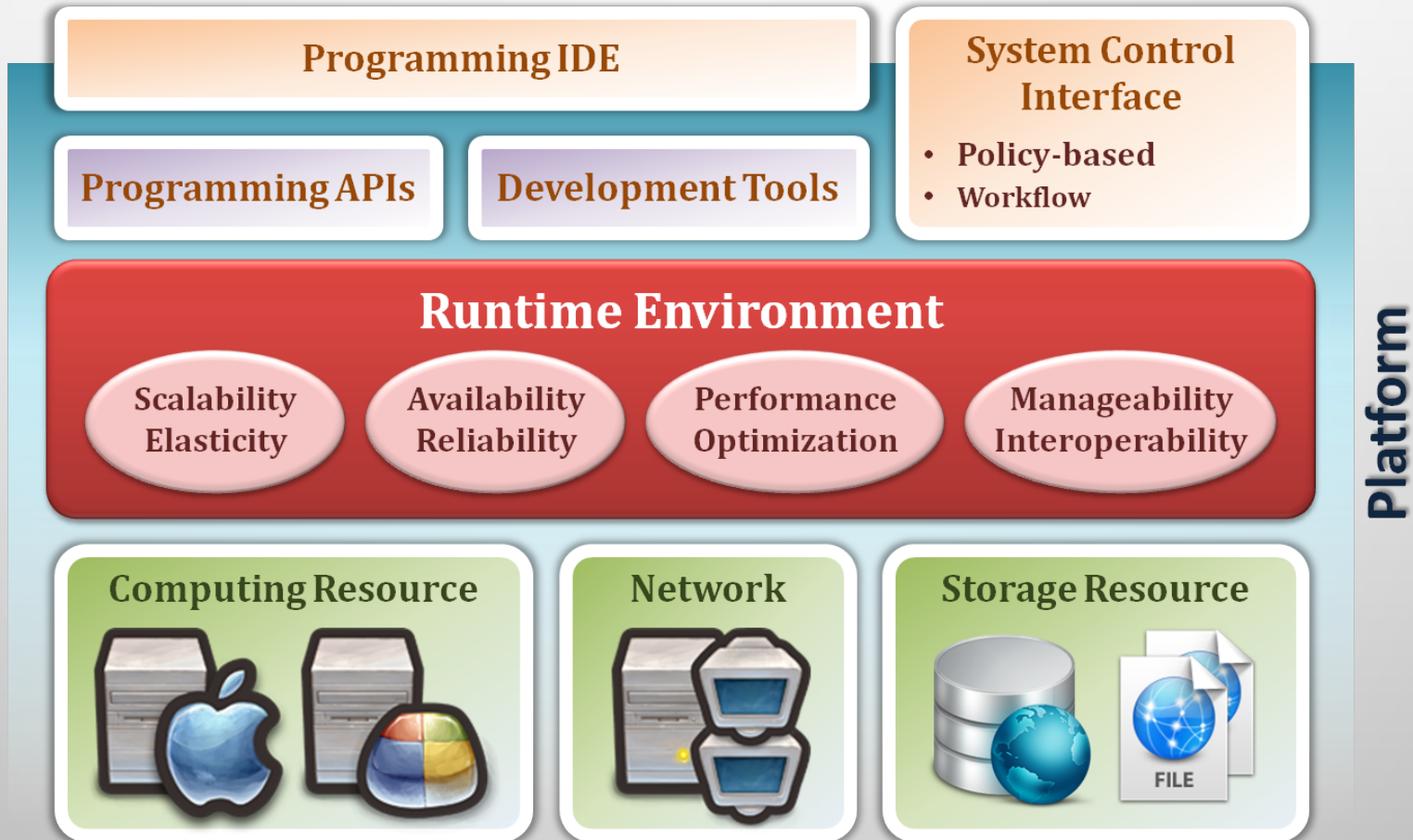


PLATFORM AS A SERVICE

- PLATFORM AS A SERVICE - PAAS
 - THE CAPABILITY PROVIDED TO THE CONSUMER IS TO DEPLOY ONTO THE CLOUD INFRASTRUCTURE CONSUMER-CREATED OR ACQUIRED APPLICATIONS CREATED USING PROGRAMMING LANGUAGES AND TOOLS SUPPORTED BY THE PROVIDER.
 - THE CONSUMER DOES NOT MANAGE OR CONTROL THE UNDERLYING CLOUD INFRASTRUCTURE INCLUDING NETWORK, SERVERS, OPERATING SYSTEMS, OR STORAGE, BUT HAS CONTROL OVER THE DEPLOYED APPLICATIONS AND POSSIBLY APPLICATION HOSTING ENVIRONMENT CONFIGURATIONS.
- EXAMPLES :
 - MICROSOFT WINDOWS AZURE
 - GOOGLE APP ENGINE
 - HADOOP
 - ... ETC

PLATFORM AS A SERVICE

- SYSTEM ARCHITECTURE :



WIKIPEDIA DEFINITIONS

- **IDE: AN INTEGRATED DEVELOPMENT ENVIRONMENT IS A SOFTWARE APPLICATION THAT PROVIDES COMPREHENSIVE FACILITIES TO COMPUTER PROGRAMMERS FOR SOFTWARE DEVELOPMENT. AN IDE NORMALLY CONSISTS OF AT LEAST A SOURCE CODE EDITOR, BUILD AUTOMATION TOOLS AND A DEBUGGER**
- **API: AN APPLICATION PROGRAMMING INTERFACE IS A COMPUTING INTERFACE TO A SOFTWARE COMPONENT OR A SYSTEM, THAT DEFINES HOW OTHER COMPONENTS OR SYSTEMS CAN USE IT.**

PLATFORM AS A SERVICE

- **ENABLING TECHNIQUE – RUNTIME ENVIRONMENT DESIGN**
 - RUNTIME ENVIRONMENT REFERS TO COLLECTION OF SOFTWARE SERVICES AVAILABLE. USUALLY IMPLEMENTED BY A COLLECTION OF PROGRAM LIBRARIES.
- **COMMON PROPERTIES IN RUNTIME ENVIRONMENT :**
 - **MANAGEABILITY AND INTEROPERABILITY**
 - **PERFORMANCE AND OPTIMIZATION**
 - **AVAILABILITY AND RELIABILITY**
 - **SCALABILITY AND ELASTICITY**



PLATFORM AS A SERVICE

- **PROVIDE SERVICE – PROGRAMMING IDE**
 - **USERS MAKE USE OF PROGRAMMING IDE (INTEGRATED DEVELOPMENT ENVIRONMENT) TO DEVELOP THEIR SERVICE AMONG PAAS.**
 - THIS IDE SHOULD INTEGRATE THE FULL FUNCTIONALITIES WHICH IS SUPPORTED FROM THE UNDERLING RUNTIME ENVIRONMENT.
 - THIS IDE SHOULD ALSO PROVIDE SOME DEVELOPMENT TOOLS, SUCH AS PROFILER, DEBUGGER AND TESTING ENVIRONMENT.
 - **THE PROGRAMMING APIS SUPPORTED FROM RUNTIME ENVIRONMENT MAY BE VARIOUS BETWEEN DIFFERENT CLOUD PROVIDERS, BUT THERE ARE STILL SOME COMMON OPERATING FUNCTIONS.**
 - COMPUTATION, STORAGE AND COMMUNICATION RESOURCE OPERATION

PLATFORM AS A SERVICE

- **PROVIDE SERVICE – SYSTEM CONTROL INTERFACE**
 - **POLICY-BASED CONTROL**
 - TYPICALLY DESCRIBED AS A PRINCIPLE OR RULE TO GUIDE DECISIONS AND ACHIEVE RATIONAL OUTCOME(S)
 - MAKE THE DECISION ACCORDING TO SOME REQUIREMENTS
 - **WORKFLOW CONTROL**
 - DESCRIBE THE FLOW OF INSTALLATION AND CONFIGURATION OF RESOURCES
 - WORKFLOW PROCESSING DAEMON DELIVERS SPEEDY AND EFFICIENT CONSTRUCTION AND MANAGEMENT OF CLOUD RESOURCES

SCI: A CONTROL INTERFACE, OR SIMPLY INTERFACE OR CONTROL PANEL, ISA MEANS BY WHICH A PERSON OPERATING A PARTICULAR PIECE OF EQUIPMENT COULD INSTRUCT THE DEVICE OR SYSTEM AS TO WHAT THEY DESIRED, AND THE ITEM BEING CONTROLLED COULD RESPOND ACCORDINGLY.

PAAS - SUMMARY

- **PAAS IS THE DEVELOPMENT PLATFORM THAT ABSTRACT THE INFRASTRUCTURE, OS, AND MIDDLEWARE TO DRIVE DEVELOPER PRODUCTIVITY.**
- PAAS ENABLING TECHNIQUE
 - RUNTIME ENVIRONMENT
- PAAS PROVIDE SERVICES
 - PROGRAMMING IDE
 - PROGRAMMING APIS
 - DEVELOPMENT TOOLS
 - SYSTEM CONTROL INTERFACE
 - POLICY BASED APPROACH
 - WORKFLOW BASED APPROACH

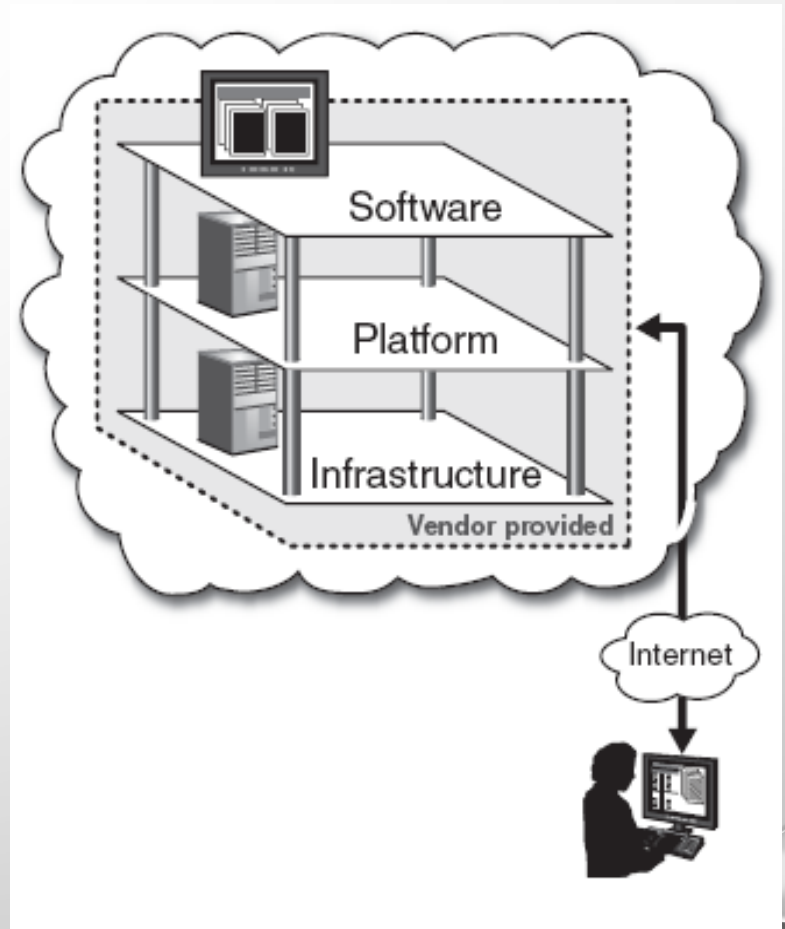
SERVICE MODELS

- INFRASTRUCTURE AS A SERVICE
- PLATFORM AS A SERVICE
- SOFTWARE AS A SERVICE

SERVICE MODELS

SOFTWARE-AS-A-SERVICE (SAAS)

- ALSO REFERRED TO AS “SOFTWARE ON DEMAND,” THIS SERVICE MODEL INVOLVES OUTSOURCING THE INFRASTRUCTURE, PLATFORM, AND SOFTWARE/APPLICATIONS.
- TYPICALLY, THESE SERVICES ARE AVAILABLE TO THE CUSTOMER FOR A FEE, PAY-AS-YOU-GO, OR A NO CHARGE MODEL.
- THE CUSTOMER ACCESSES THE APPLICATIONS OVER THE INTERNET.



SOFTWARE AS A SERVICE

- SOFTWARE AS A SERVICE - SAAS
 - THE CAPABILITY PROVIDED TO THE CONSUMER IS TO USE THE PROVIDER'S APPLICATIONS RUNNING ON A CLOUD INFRASTRUCTURE. THE APPLICATIONS ARE ACCESSIBLE FROM VARIOUS CLIENT DEVICES THROUGH A THIN CLIENT INTERFACE SUCH AS A WEB BROWSER (E.G., WEB-BASED EMAIL).
 - THE CONSUMER DOES NOT MANAGE OR CONTROL THE UNDERLYING CLOUD INFRASTRUCTURE INCLUDING NETWORK, SERVERS, OPERATING SYSTEMS, STORAGE, OR EVEN INDIVIDUAL APPLICATION CAPABILITIES, WITH THE POSSIBLE EXCEPTION OF LIMITED USER-SPECIFIC APPLICATION CONFIGURATION SETTINGS.
- EXAMPLES :
 - GOOGLE APPS (E.G., GMAIL, GOOGLE DOCS, GOOGLE SITES, ...ETC)
 - SALESFORCE.COM
 - EYEOS
 - ... ETC

SOFTWARE AS A SERVICE



SOFTWARE AS A SERVICE

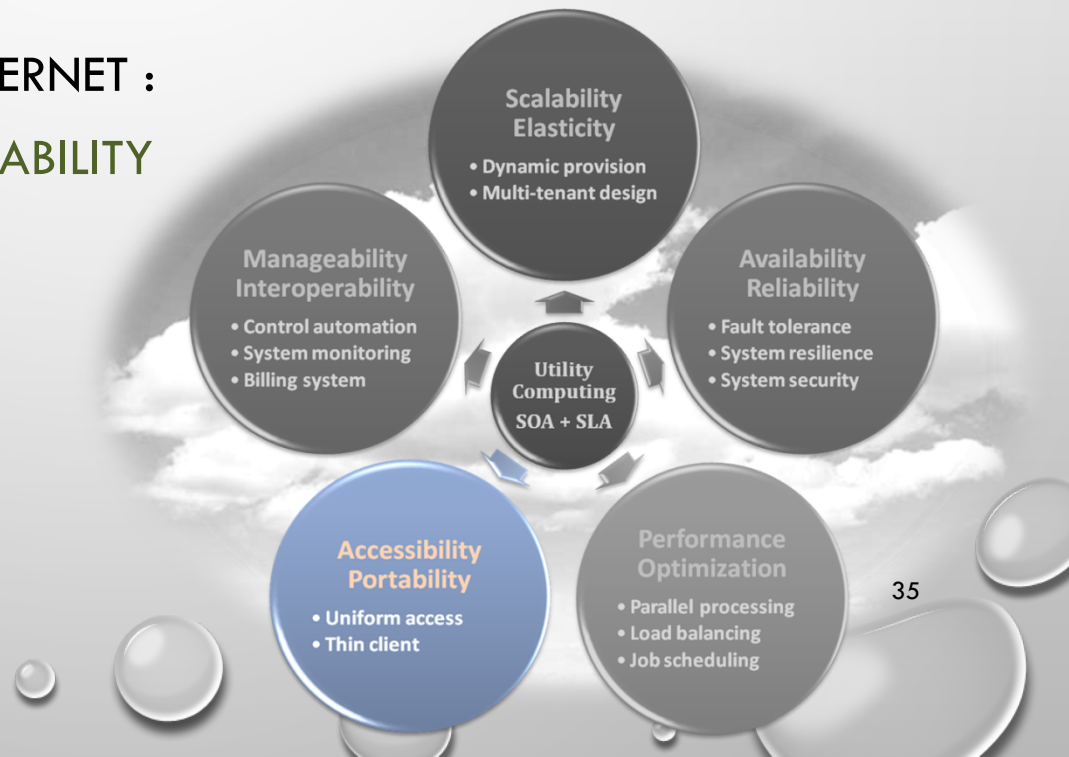
- **ENABLING TECHNIQUE – WEB SERVICE**

- **WEB 2.0 IS THE TREND OF USING THE FULL POTENTIAL OF THE WEB**

- VIEWING THE INTERNET AS A COMPUTING PLATFORM
- RUNNING INTERACTIVE APPLICATIONS THROUGH A WEB BROWSER
- LEVERAGING INTERCONNECTIVITY AND MOBILITY OF DEVICES
- ENHANCED EFFECTIVENESS WITH GREATER HUMAN PARTICIPATION

- **PROPERTIES PROVIDED BY INTERNET :**

- **ACCESSIBILITY AND PORTABILITY**



SOFTWARE AS A SERVICE

- PROVIDE SERVICE – **WEB-BASED APPLICATIONS**
 - CONVENTIONAL APPLICATIONS SHOULD TRANSLATE THEIR ACCESS INTERFACE ONTO WEB-BASED PLATFORM.
 - APPLICATIONS IN DIFFERENT DOMAINS
 - **GENERAL APPLICATIONS** – APPLICATIONS WHICH ARE DESIGNED FOR GENERAL PROPOSE, SUCH AS **OFFICE SUIT**, **MULTIMEDIA** AND **INSTANT MESSAGE**, ...ETC.
 - **BUSINESS APPLICATIONS** – APPLICATION WHICH ARE DESIGNED FOR BUSINESS PROPOSE, SUCH AS **ERP**, **CRM** AND **MARKET TRADING SYSTEM**, ...ETC.
 - **SCIENTIFIC APPLICATIONS** – APPLICATION WHICH ARE DESIGNED FOR SCIENTIFIC PROPOSE, SUCH AS **AEROSPACE SIMULATION** AND **BIOCHEMISTRY SIMULATION**, ... ETC.
 - **GOVERNMENT APPLICATIONS** – APPLICATIONS WHICH ARE DESIGNED FOR GOVERNMENT PROPOSE, SUCH AS **NATIONAL MEDICAL SYSTEM** AND **PUBLIC TRANSPORTATION SYSTEM SERVICE**, ...ETC.

SOFTWARE AS A SERVICE

- **PROVIDE SERVICE – WEB PORTAL**

- APART FROM THE STANDARD SEARCH ENGINE FEATURE, WEB PORTALS OFFER OTHER SERVICES SUCH AS E-MAIL, NEWS, STOCK PRICES, INFORMATION, DATABASES AND ENTERTAINMENT.
- PORTALS PROVIDE A WAY FOR ENTERPRISES TO PROVIDE A CONSISTENT LOOK AND FEEL WITH ACCESS CONTROL AND PROCEDURES FOR MULTIPLE APPLICATIONS AND DATABASES, WHICH OTHERWISE WOULD HAVE BEEN DIFFERENT ENTITIES ALTOGETHER.
- SOME EXAMPLES :
 - IGOOGLE
 - MSNBC
 - NETVIBES
 - YAHOO!

SAAS - SUMMARY

- **SAAS IS THE FINISHED APPLICATIONS THAT YOU RENT AND CUSTOMIZE.**
- SAAS ENABLING TECHNIQUE
 - WEB SERVICE
- SAAS PROVIDE SERVICES
 - WEB-BASED APPLICATIONS
 - GENERAL APPLICATIONS
 - BUSINESS APPLICATIONS
 - SCIENTIFIC APPLICATIONS
 - GOVERNMENT APPLICATIONS
 - WEB PORTAL

DEPLOYMENT MODELS

- HOW TO DEPLOY A CLOUD SYSTEM ?



DEPLOYMENT MODEL

- THERE ARE FOUR PRIMARY CLOUD DEPLOYMENT MODELS :
 - PUBLIC CLOUD
 - PRIVATE CLOUD
 - COMMUNITY CLOUD
 - HYBRID CLOUD
- EACH CAN EXHIBIT THE PREVIOUSLY DISCUSSED CHARACTERISTICS; THEIR DIFFERENCES LIE PRIMARILY IN THE SCOPE AND ACCESS OF PUBLISHED CLOUD SERVICES, AS THEY ARE MADE AVAILABLE TO SERVICE CONSUMERS.

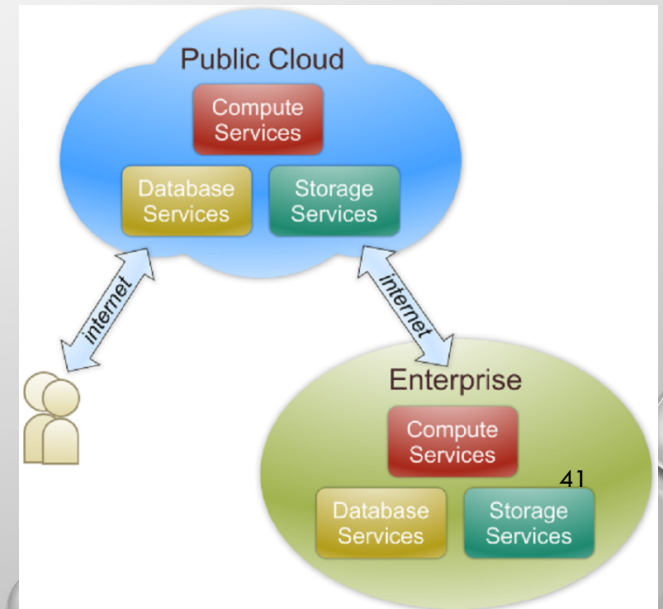
PUBLIC CLOUD

- PUBLIC CLOUD DEFINITION

- THE CLOUD INFRASTRUCTURE IS MADE AVAILABLE TO THE GENERAL PUBLIC OR A LARGE INDUSTRY GROUP AND IS OWNED BY AN ORGANIZATION SELLING CLOUD SERVICES.
- ALSO KNOWN AS EXTERNAL CLOUD OR MULTI-TENANT CLOUD, THIS MODEL ESSENTIALLY REPRESENTS A CLOUD ENVIRONMENT THAT IS OPENLY ACCESSIBLE.

- BASIC CHARACTERISTICS :

- HOMOGENEOUS INFRASTRUCTURE
- COMMON POLICIES
- SHARED RESOURCES AND MULTI-TENANT
- LEASED OR RENTED INFRASTRUCTURE
- ECONOMIES OF SCALE



PUBLIC CLOUD

- AMAZON ELASTIC COMPUTE **CLOUD** (EC2)
- IBM'S BLUE **CLOUD**
- SUN **CLOUD**
- GOOGLE APPENGINE
- GOOGLE DRIVE
- DROPBOX
- WINDOWS AZURE SERVICES PLATFORM
-

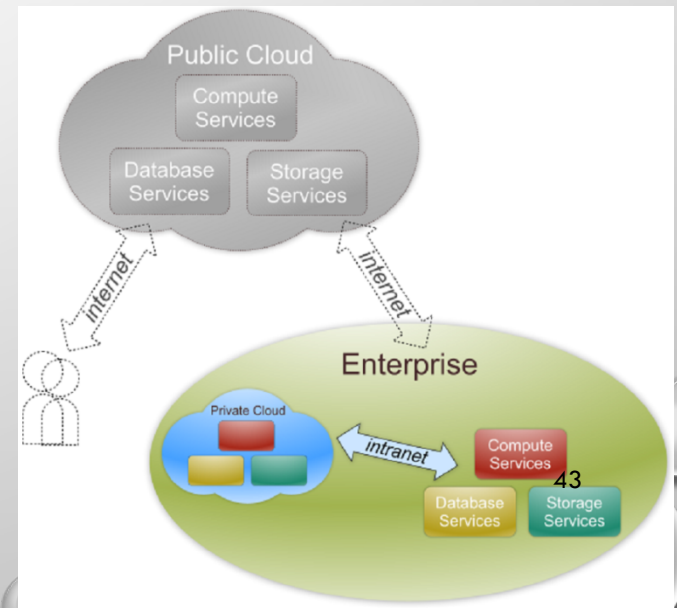
PRIVATE CLOUD

- PRIVATE CLOUD DEFINITION

- THE CLOUD INFRASTRUCTURE IS OPERATED SOLELY FOR AN ORGANIZATION. IT MAY BE MANAGED BY THE ORGANIZATION OR A THIRD PARTY AND MAY EXIST ON PREMISE OR OFF PREMISE.
- ALSO REFERRED TO AS INTERNAL CLOUD OR ON-PREMISE CLOUD, A PRIVATE CLOUD INTENTIONALLY LIMITS ACCESS TO ITS RESOURCES TO SERVICE CONSUMERS THAT BELONG TO THE SAME ORGANIZATION THAT OWNS THE CLOUD.

- BASIC CHARACTERISTICS :

- HETEROGENEOUS INFRASTRUCTURE
- CUSTOMIZED AND TAILORED POLICIES
- DEDICATED RESOURCES
- IN-HOUSE INFRASTRUCTURE
- END-TO-END CONTROL



PRIVATE CLOUD

- EUCALYPTUS.
- UBUNTU ENTERPRISE **CLOUD** - UEC (POWERED BY EUCALYPTUS)
- AMAZON VPC (VIRTUAL PRIVATE **CLOUD**)
- VMWARE **CLOUD** INFRASTRUCTURE SUITE.
- MICROSOFT ECI DATA CENTER.
-

PUBLIC VS. PRIVATE

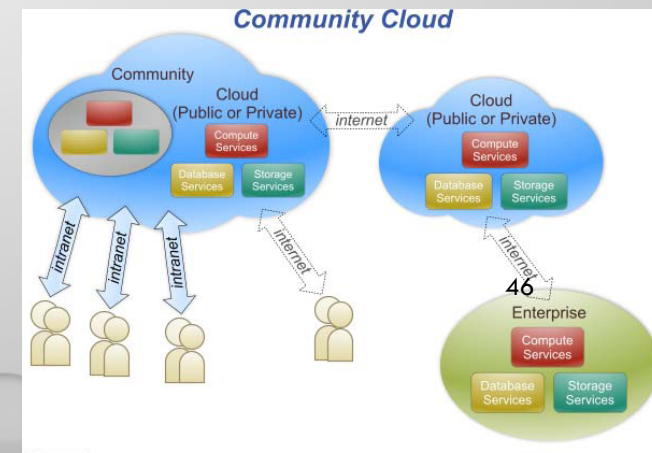
- COMPARISON :

	Public Cloud	Private Cloud
<i>Infrastructure</i>	<i>Homogeneous</i>	<i>Heterogeneous</i>
<i>Policy Model</i>	<i>Common defined</i>	<i>Customized & Tailored</i>
<i>Resource Model</i>	<i>Shared & Multi-tenant</i>	<i>Dedicated</i>
<i>Cost Model</i>	<i>Operational expenditure</i>	<i>Capital expenditure</i>
<i>Economy Model</i>	<i>Large economy of scale</i>	<i>End-to-end control</i>

COMMUNITY CLOUD

- COMMUNITY CLOUD DEFINITION

- THE CLOUD INFRASTRUCTURE IS SHARED BY SEVERAL ORGANIZATIONS AND SUPPORTS A SPECIFIC COMMUNITY THAT HAS SHARED CONCERNS (E.G., MISSION, SECURITY REQUIREMENTS, POLICY, AND COMPLIANCE CONSIDERATIONS).
- A **COMMUNITY CLOUD** IS A **CLOUD** SERVICE MODEL THAT PROVIDES A **CLOUD COMPUTING** SOLUTION TO A LIMITED NUMBER OF INDIVIDUALS OR ORGANIZATIONS THAT IS GOVERNED, MANAGED AND SECURED COMMONLY BY ALL THE PARTICIPATING ORGANIZATIONS OR A THIRD PARTY MANAGED SERVICE PROVIDER.



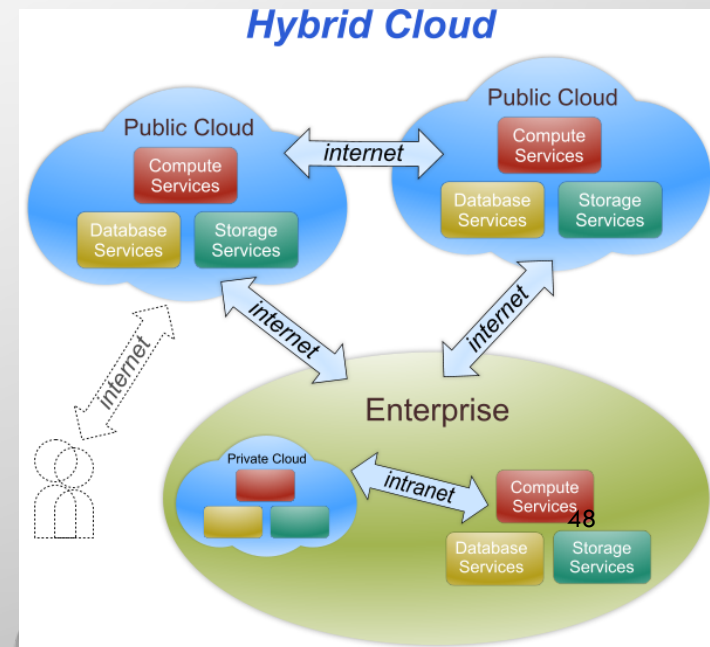
COMMUNITY CLOUD

- INTER UNIVERSITY RESEARCH CLUSTER
- INTERPOL
-

HYBRID CLOUD

- HYBRID CLOUD DEFINITION

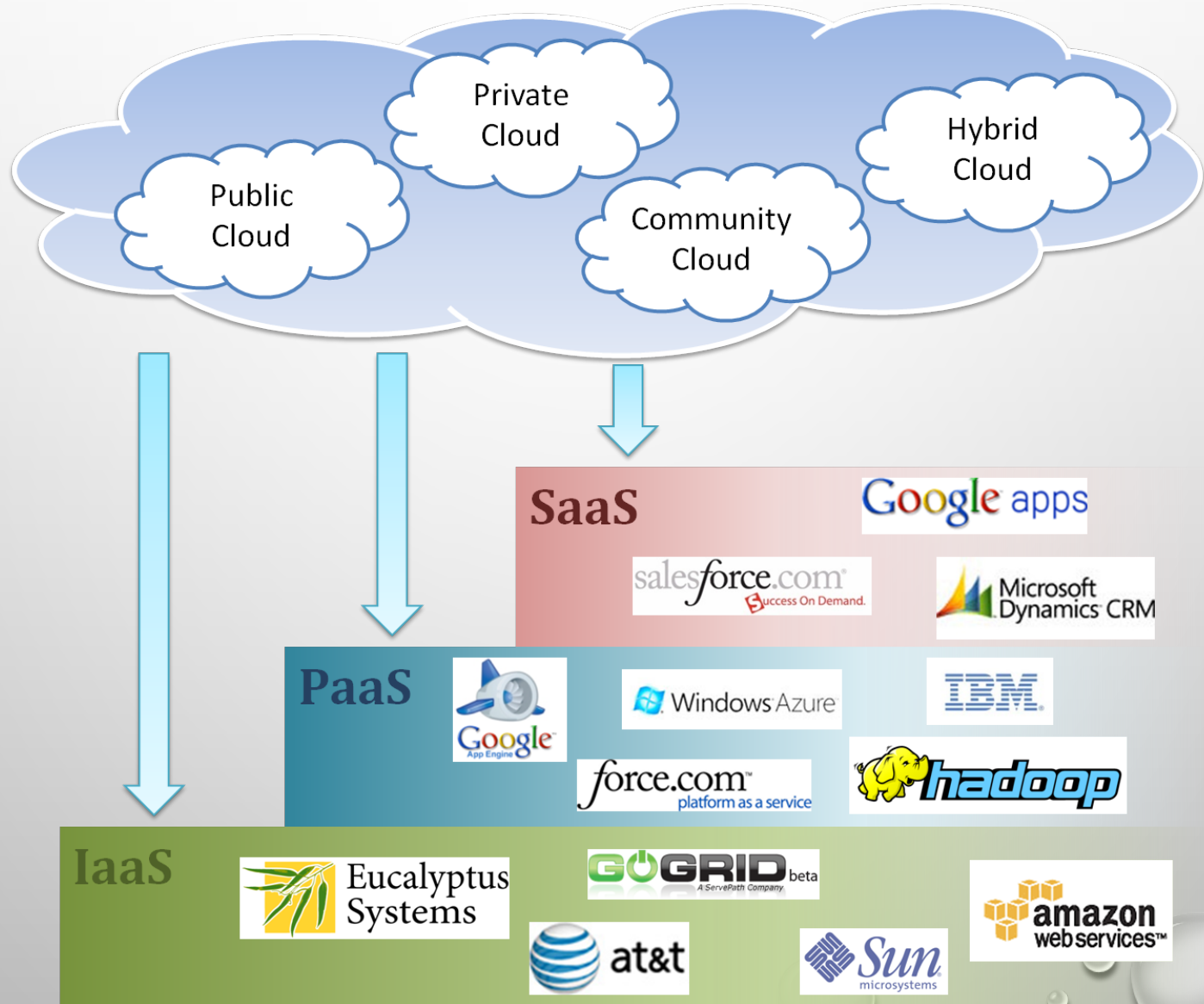
- THE CLOUD INFRASTRUCTURE IS A COMPOSITION OF TWO OR MORE CLOUDS (PRIVATE, COMMUNITY, OR PUBLIC) THAT REMAIN UNIQUE ENTITIES BUT ARE BOUND TOGETHER BY STANDARDIZED OR PROPRIETARY TECHNOLOGY THAT ENABLES DATA AND APPLICATION PORTABILITY (E.G., CLOUD BURSTING FOR LOAD-BALANCING BETWEEN CLOUDS).



HYBRID CLOUD

- VMWARE IS PRIVATE CLOUD AND AWS, GOOGLE ARE PUBLIC CLOUDS.
- HYBRID CLOUD IS A SITUATION WHERE COMPANIES USE BOTH PRIVATE AND PUBLIC CLOUDS FOR THEIR BUSINESS NEEDS.

CLOUD ECOSYSTEM



SUMMARY

- WHAT IS CLOUD COMPUTING IN YOUR MIND
 - CLEAR OR CLOUDY?
- CLOUD COMPUTING IS A NEW PARADIGM SHIFT OF COMPUTING
- CLOUD COMPUTING CAN PROVIDE HIGH QUALITY OF PROPERTIES AND CHARACTERISTICS BASED ON ESSENTIALLY CENTRAL IDEAS
- SERVICE MODELS AND DEPLOYMENT MODELS PROVIDE SERVICES THAT CAN BE USED TO
 - RENT FUNDAMENTAL COMPUTING RESOURCES
 - DEPLOY AND DEVELOP CUSTOMER-CREATED APPLICATIONS ON CLOUDS
 - ACCESS PROVIDER'S APPLICATIONS OVER NETWORK (WIRED OR WIRELESS)



雲端計算 *CLOUD COMPUTING*

- INTRODUCTION TO CLOUD COMPUTING

REFERENCES

- NIST (National Institute of Standards and Technology). <http://csrc.nist.gov/groups/SNS/cloud-computing/>
- M. Armbrust et. al., “Above the Clouds: A Berkeley View of Cloud Computing,” Technical Report No. UCB/EECS-2009-28, University of California at Berkeley, 2009.
- R. Buyya et. al., “Cloud computing and emerging IT platforms: Vision, hype, and reality for delivering computing as the 5th utility,” Future Generation Computer Systems, 2009.
- Cloud Computing Use Cases. <http://groups.google.com/group/cloud-computing-use-cases>
- Cloud Computing Explained. <http://www.andyharjanto.com/2009/11/wanted-cloud-computing-explained-in.html>
- From Wikipedia, the free encyclopedia
- All resources of the materials and pictures were partially retrieved from the Internet.

